

Reduced management activities: Tasks such as storage maintenance, adding new storage systems to an existing cluster, configuring RAID's, etc, are all offloaded to the cloud storage provider.

Besides these, cloud storage also provides built-in multi-tenancy security in the form of encryption and checksum facilities, easier data transfers between on-premise storage systems, and the cloud using reliable upload mechanisms such as multi-part uploads, etc.

A few noted cloud storage providers and their storage

service offerings are mentioned below.

There are a lot of open source platforms available in the market that you can use to build your own cloud storage system, each with their own pros and cons. Given below is a quick comparison between some of the most widely used cloud storage platforms: Basho's Riak CS, Openstack Swift, Inktank's Ceph and Walrus from Eucalyptus.

Introduction to Riak CS

Riak CS (cloud storage) is an open source platform

Cloud storage provider	Name of the service	URL
Amazon Web Services (AWS)	Simple Storage Service (S3)	http://aws.amazon.com/s3/
Rackspace	Cloud Files Services	http://www.rackspace.com/cloud/files/
Google	Google Cloud Storage	https://cloud.google.com/products/cloud-storage
Windows Azure	Windows Azure Blog Storage	http://www.windowsazure.com/en-us/solutions/storage-backup-recovery/
HP	HP Cloud Object Storage	http://www.hpcloud.com/products-services/object-storage
IBM	IBM SmartCloud Object Storage	http://www-935.ibm.com/services/in/en/cloud-enterprise/object-storage.html

	Riak CS	Swift	Ceph	Walrus
What is it?	Open source distributed cloud storage platform that is easy to scale and highly available.	A distributed object storage system optimised for multi-tenancy and high concurrency.	A software storage platform that provides object, block and file storage from a single distributed computer cluster.	A persistent 'storage as a service' feature provided by Eucalyptus that can be used with simple HTTP put/get commands.
Installation	Standalone installation.	Though it can run on its own, Swift is part of the OpenStack project.	Standalone installation.	Requires additional Eucalyptus Cloud components to be installed along with it.
Languages	Erlang	Python	C/ C++	Java
Features	Durability, scalability, high availability, fault tolerant	Durability, scalability, high availability, fault tolerant	Scalability, high availability, fault tolerant	Scalability, high availability
Support for Amazon S3	Directly and natively supports Amazon S3 APIs. Provides a rich set of native clients as well to interact with various cloud storage services.	Comes equipped with its own flavour of storage (non-S3 based) API, with its own strengths. Support for the Amazon S3 API is optionally provided via a community-backed middleware that sits on top of the Swift framework.	Provides Amazon S3 and Swift compatible interfaces.	Provides an S3 compatible API for object storage, which enables easy storage and retrieval of persistent data.
Multi-tenancy support	Supported	Supported	Limited	Limited
Multi-datacentre replication	Supported	Supported	Limited	Limited
References	http://basho.com/riak-cloud-storage/	http://docs.openstack.org/developer/swift/	http://ceph.com/docs/master/	https://github.com/eucalyptus/eucalyptus/wiki/Walrus